State AI Policy: How Is Regulation Responding to the Rise of AI?

SIADS 696 Milestone II Project Report

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1. Introduction

State lawmakers introduced more than 1,000 AI measures in 2025, addressing transparency, health care, and chatbots. As artificial intelligence continues to expand across government, business, and consumer applications, states are increasingly developing their own approaches to AI governance. California's September 2026 AI oversight laws show why comparing state priorities and bill outcomes remains timely. Comparing state bills can show which concerns appear in legislation and how approaches differ across jurisdictions.

We ask two related questions: What themes appear in state AI bills, and what information helps distinguish their recorded outcomes? We use nonnegative matrix factorization (NMF) and latent Dirichlet allocation (LDA) to explore the themes in bill summaries. We also compare classifiers that use bill details, such as state and sponsor information, along with text and topic features. Because some text and metadata reflect later stages of a bill, this is a retrospective analysis. The results describe patterns in recorded outcomes. They do not show that a feature caused an outcome or provide predictions made when a bill was introduced.

On the original 2026 test, the selected support vector machine (SVM) using bill details correctly classified 53.0% of bills. A simpler model that labeled every bill Pending was correct 76.2% of the time because Pending was the most common outcome. Accuracy alone does not tell the whole story: the SVM had a higher macro F1 score, 0.442 compared with 0.288. Macro F1 combines how often the model is right when it predicts an outcome with what share of bills in that group it finds. It averages those scores across the three outcomes; a score closer to 1 is better. The SVM also classified 283 bills as Failed even though they were recorded as Pending.

The topic models found recurring language about synthetic media, automated decisions, health care and pricing. Two NMF topics were dominated by general AI wording, however, so not every topic represented a distinct policy area. Examining eight selected bills also showed that measures in different areas can share approaches, such as requiring human involvement. The topic models help readers navigate the collection, but adding topic features did not consistently improve outcome classification in one comparison across years. The classification errors show why model-generated statuses should be checked against official records.

2. Related Work

Bhargava and Soliman's (2026) State AI Legislation Corpus and its accompanying Bill_Analyst_Agent system provide our source records and the closest point of comparison. Their system uses a multi-agent language-model pipeline to assign bills to seven predefined policy dimensions. We use the published records to ask different questions: NMF and LDA learn themes from the summaries, while separate classifiers distinguish recorded outcomes. We do not reproduce the publisher's classification pipeline or treat its taxonomy scores as human-verified labels. We label our topics by reviewing model terms and representative summaries.

Nay (2017) combines bill text and contextual metadata to study legislative outcomes. The study trains on earlier U.S. Congresses and evaluates later ones. It also compares information available at introduction with the newest text and metadata. This work motivates our comparisons of metadata and text, as well as our attention to when information becomes available. Our setting differs in scale, jurisdiction and target: we study state AI measures, retain an unresolved Pending class, and use a published snapshot. Our scores therefore cannot be compared directly with Nay's enactment forecasts.

Quinn et al. (2010) infer topics from U.S. Senate speeches to study political attention without defining categories in advance. Their work provides a comparison for interpreting learned policy themes. We analyze state bill summaries, compare NMF with LDA, and examine how topic weights vary across states. A high topic weight describes language represented in this corpus; it does not by itself establish a state's policy priorities or an issue's importance.

3. Data Sources

Our primary source is Bhargava and Soliman's U.S. State AI Legislation Corpus, version 4, published on Zenodo on June 21, 2026. Its public 51.8 MB file, Final_Database_10.csv, contains 2,571 records and 29 columns. The corpus combines National Conference of State Legislatures (NCSL) records, LegiScan sponsor metadata and publisher-generated policy classifications. Fields include bill text, summaries, state, introduction and text-version dates, sponsor affiliations, and recorded status. We preserve the downloaded file as data/raw/Final_Database.csv. Although the publisher describes coverage as 2019–2026, our parsed introduction dates contain one 2018 record. That record is excluded from the predefined supervised year splits.

Shared preparation standardizes Unicode and whitespace, parses available dates, and adds a source-row locator and eight sponsor features. The prepared table contains 2,567 rows and 45 columns. We remove four rows with identical cleaned bill text, keeping the last occurrence in source order. This removes exact text repetitions but does not establish that every remaining record is a unique bill or that the retained metadata are the most accurate. The original CSV remains unchanged. The preparation script and notebook document these transformations so both analyses begin with the same table.

Table 1 summarizes the publisher's Bill_Status_Grouped labels used by the classifiers:

Table 1. Recorded outcome groups in the prepared dataset.

Group	Prepared records	Source statuses included
Failed	764	Failed and Vetoed
Passed	323	Enacted and Adopted
Pending	1,480	Pending, To Governor and Override Pending

These groups describe the source snapshot. Pending includes unresolved bills, and Passed does not mean that every record is an enacted law. The lowercase status column tracks publisher processing, not legislative outcomes.

The prepared table contains bill text and NCSL descriptions for every record, but 484 records lack both the separate summary and extract fields. We kept them because a missing summary does not mean the bill text is missing. Our topic models use NCSL descriptions, so those missing summaries do not remove records from the topic analysis. Each supervised comparison explains which text it uses and any records it filters out. The source_row column points to a record's position in the original CSV; it is not a unique bill ID or a model input.

We checked selected records against their sources and reviewed the histories of pre-2026 bills retrieved through the LegiScan API on October 25, 2026. The larger review matched 1,523 ordinary bills and supported changing 39 labels from Pending to Passed based on dated outcomes. Eighteen of those outcomes were recorded by the December 31, 2024 training cutoff, and 21 by the December 31, 2025 evaluation cutoff. We did not label a veto as a resolved failure if further consideration was still pending and no dated closure was documented by the cutoff. These observations remain separate from the original snapshot labels used for the

final test. We also found 30 year-boundary differences involving filing or pre-session milestones; we kept the original dates rather than automatically replacing formal introduction dates.

4. Feature Engineering

The primary objective in our feature engineering is to convert the sponsor metadata into structured features for our classification models. The original dataset contained two sponsor related variables Sponsor and Sponsor Party. Since each bill could have multiple sponsors, the values on these features were lists of sponsor names and political parties of each sponsor.

We focused on the Sponsor Political party to derive numerical and binary features representing the number of sponsors, political party composition and bipartisan participation. These features were selected to investigate whether sponsorship patterns are associated with legislative status.

Table 2. Sponsor features used in the classifiers.

Feature	Description
Total_Sponsors	Count of all sponsors in list
D_Count	Count of all Democrat sponsors
R_Count	Count of all Republican sponsors
Other_Count	Count of all other known political affiliations
Unknown_Count	Missing party affiliation
Bipartisan_Support	D and R sponsor present = 1, Any other case of party representation = 0
Democrat_Sponsor_Ratio	D_Count/Total_Sponsors
Republican_Sponsor_Ratio	R_Count/Total_Sponsors

The sponsor counts capture the composition and size of a bill's sponsorship group, while the ratio features normalized party representation relative to the total number of sponsors. This distinction allows the models to differentiate between the absolute number of sponsors and the relative political composition of sponsorship.

Additionally, the Bipartisan_Support feature was introduced to identify bills with sponsorship from both major political parties. This feature allows us to investigate whether bipartisan participation provides additional information for distinguishing legislative statuses.

These eight engineered features were incorporated into the baseline classification models alongside other legislative metadata, including jurisdiction, measure code, and state political control. They were also retained as candidate predictors during subsequent model development.

5. Supervised Learning

Methods

The supervised learning component in this project starts from a baseline of 3 classification algorithms to identify patterns associated with the recorded status of AI-related bills: Passed, Failed and Pending.

The main goal is to classify the bill status retrospectively using the available data, rather than forecast the outcome of newly introduced legislation. This is an important distinction as the dataset contains information that may have been updated throughout the life of the Bill.
We compared three algorithms to compare different approaches to learning from structured metadata and textual information.

- Logistic Regression: Simple and interpretable baseline to compare against other models.
- Linear Support Vector Machine: Better performance across sparse matrices, supporting our text features and metadata.
- XGBoost: Enrich complex patterns between text features and metadata.

We created a baseline for each of these methods with three data subsets:

- Metadata: Sponsor data, Jurisdiction, measure code, state political control, engineered sponsor features
- Metadata + NCSL_Summary: Metadata combined with summaries provided by the National Conference of State Legislatures. .
- Metadata + Bill_Text: Metadata combined with full bill text

Categorical metadata was transformed using one-hot encoding, while numerical features were imputed and standardized where appropriate for the selected algorithm. Textual features were transformed using Term Frequency–Inverse Document Frequency (TF-IDF), allowing the models to represent legislative language numerically.

All models were evaluated primarily using macro-averaged F1, which calculates the F1 score independently for each class and averages the results with equal weight. This metric was selected because our dataset contains an unequal distribution of Passed, Failed, and Pending bills, and overall accuracy could obscure poor performance on minority classes.

We also evaluated macro precision, macro recall, weighted F1, balanced accuracy, and class-specific precision and recall for Passed legislation. These additional metrics provide insight into the models' ability to distinguish between legislative statuses, particularly the less frequently represented Passed class.

Data Partitions

The main challenge in this project was defining the right training, validation and test splits.
The NCSL dataset contains legislative records reflecting information available at the time of data collection. However, bills evolve throughout the legislative process, and their metadata and textual content may change between introduction, amendment, and final legislative action, and these changes were tracked as updates to individual bills and no historical data was available. This introduced a source of temporal data leakage, as bills in later versions entered terminology or metadata associated with their status.

To review generalization across legislative years, we decided to perform a chronological split as below:

Train 2019 - 2024 bills	Validation 2025 Bills	Test 2026 Bills
Total Bills 722	Total Bills 1016	Total Bills 828
Failed 45.3%	Pending 54.1%	Pending 76.2%
Pending 41.4%	Failed 27.7%	Failed 18.8%
Passed 13.3%	Passed 18.2%	Passed 5.0%

Table 3. Bill outcomes by training, validation and test split.

This split allowed us to test models trained on earlier legislation against records from subsequent years to have a more realistic assessment of the model's generalization. This did not fully remove the risk of data leakage as bills continued to have verbiage from later versions and has been identified as an opportunity for future data collection and model refinement.

An additional challenge was the variation in class distributions across legislative years. In particular, the 2026 test partition contained a substantially larger proportion of Pending legislation than the training and validation partitions. This is expected on this partition as bills introduced in 2026 could be early in the legislation process and therefore may not have reached a final outcome, however it highlights the potential for minority class misclassification.

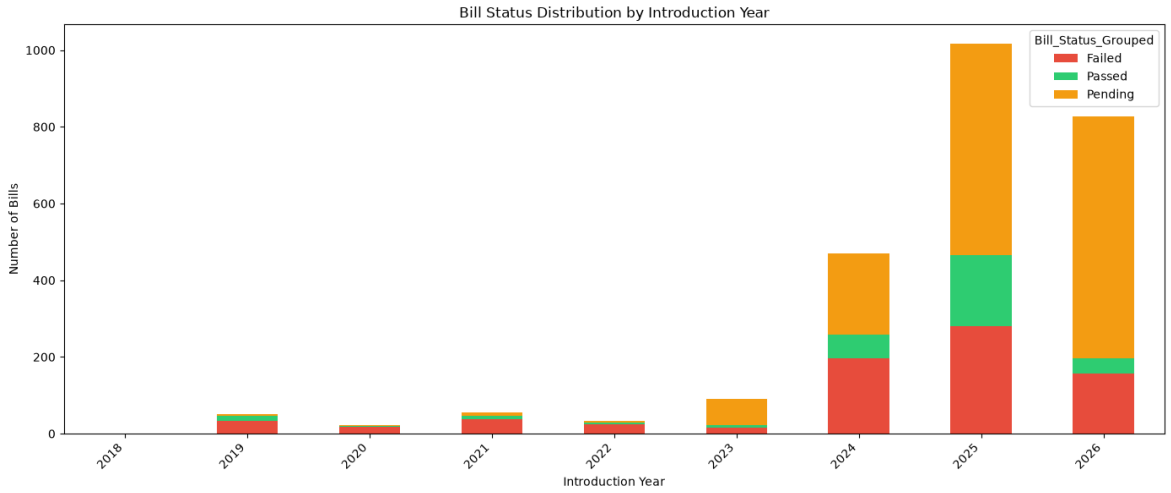


Figure 1. Recorded bill outcomes by introduction year.

Baseline Model Results

The baseline experiments showed us important differences in performance across models and feature configurations.

Linear SVM achieved the strongest validation performance across all three feature configurations. The metadata-only model achieved a macro f1 score of .5393, increasing to .5508 with bill summaries and to .5659 with full bill text. This suggests that textual information could be providing additional classification value.

Logistic regression demonstrated a relatively stable validation performance across all configurations, with Macro F1 scores ranging from 0.4903 to 0.4970. Although we see a slight improvement after adding text to the model, the increase was smaller than the Linear SVM.

XGBoost showed us the greatest discrepancy between training and validation performance. Its metadata and full text model achieved a training macro F1 score of .9979, but a validation macro F1 score of 0.4475. This is evidence that the model did great at fitting the training data but was unable to generalize the legislation at the validation year, proving its tendency to overfit at this baseline. Further evaluation is taken with parameter fine tuning to reduce this tendency.

Although SVM's strong performance suggested it to be the best model, identifying Passed legislation remained a challenge for all models. The best performing SVM model achieved a Passed recall of 0.2378, indicating that only about 24% of Passed bills were correctly identified.

These findings suggest that model complexity may not be enough to improve legislative status classification. The results highlight the importance of data quality, class imbalance and feature representation, as these could improve our overall model performance.

Model	Inputs	Train macro F1	Validation macro F1	Validation macro precision	Validation macro recall	Passed precision	Passed recall
Linear SVM	Metadata + text	0.9657	0.5659	0.6263	0.5889	0.4835	0.2378
Linear SVM	Metadata + summary	0.9767	0.5508	0.6192	0.5761	0.4824	0.2216
Linear SVM	Metadata	0.6982	0.5393	0.5948	0.5644	0.4023	0.1892
Logistic reg.	Metadata + text	0.8017	0.4970	0.5880	0.5483	0.4054	0.0811
Logistic reg.	Metadata + summary	0.8172	0.4949	0.6091	0.5488	0.4688	0.0811
Logistic reg.	Metadata	0.6908	0.4903	0.5534	0.5370	0.2941	0.0811
XGBoost	Metadata	0.7327	0.4749	0.5275	0.5193	0.2766	0.0703
XGBoost	Metadata + summary	0.8876	0.4653	0.5985	0.5232	0.5294	0.0486
XGBoost	Metadata + text	0.9979	0.4475	0.6584	0.5174	0.6400	0.0865

Table 4. Baseline model results on the 2025 validation set.

Model Selection and Evaluation

In order to improve our baselines we proceeded to evaluate additional model configurations to improve generalization and minority-class classification. We introduced balanced class weights for Logistic Regression, and Linear SVM, stronger regularization for XGBoost, and additional topic-based features derived from our unsupervised learning exercise (NMF and LDA).
Model selection was created by three expanding folds: train on 2019–2021 and validate on 2022, then train through 2022 and validate on 2023, then train through 2023 and validate on 2024. The 2025 comparison trains in 2019–2024. Each configuration uses the same rows in a cohort.

The results highlighted tradeoffs between model complexity, predictive performance, and consistency across legislative years. Although adding full bill text improved Linear SVM's 2025 validation macro F1 from 0.5701 to 0.5770 under balanced settings, the metadata-only model achieved a higher average macro F1 across historical validation years (0.4860 versus 0.4060). Furthermore, the metadata-only model used substantially fewer transformed features (79 versus 5,079) and demonstrated a smaller training-validation performance gap. These findings supported selecting Linear SVM with metadata features, C=10, and balanced class weights as our final classifier, prioritizing simplicity and more consistent historical performance over a small improvement in a single validation year.

The chart below shows the comparison in model families with the new configuration as compared to the original baseline.

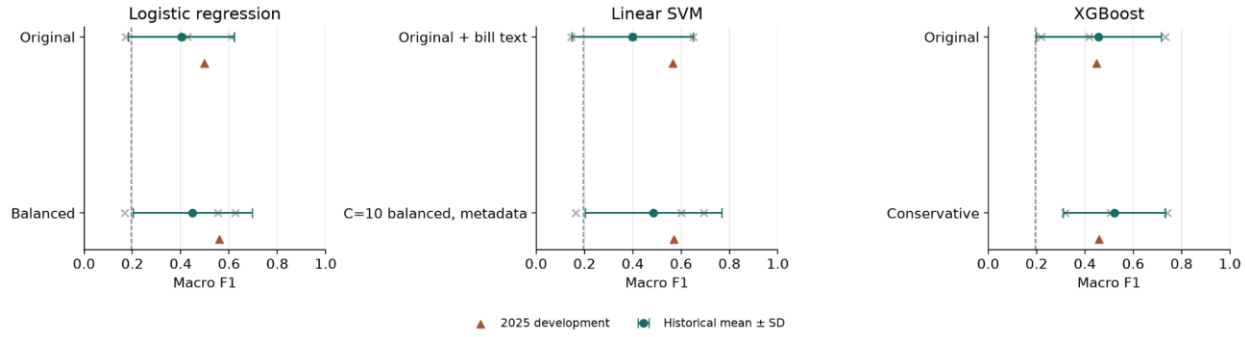


Figure 2. Macro F1 across historical validation years and 2025 for the model comparisons.

After model selection, the final classifier was retrained using 1,738 legislative records from 2019–2025 and evaluated on the held-out 2026 dataset containing 828 records. The model achieved a macro F1 score of 0.4422 and balanced accuracy of 0.5360, outperforming a majority-class reference model with scores of 0.2883 and 0.3333, respectively. However, its overall accuracy of 53.02% was lower than the majority reference's 76.21%, reflecting the model's greater emphasis on identifying minority outcomes rather than predicting the dominant Pending class.

At the class level, the final model achieved a recall of 0.6667 for Failed, 0.4390 for Passed, and 0.5024 for Pending legislation. Although the model improved minority-class coverage relative to the majority-class reference, its lower performance in 2026 compared with the 2025 development results highlights the difficulty of generalizing across legislative years with changing status distributions.

Overall, the evaluation demonstrates that a simpler metadata-based classifier offers a reasonable balance between model complexity and retrospective status classification performance. Nevertheless, class imbalance, changing legislative outcomes, and potential temporal leakage remain important limitations. The results should therefore be interpreted as

classification of recorded legislative statuses rather than reliable forecasts of eventual bill passage.

Failure Analysis

Our selected model excludes the text features, reducing exposure to potential data leakage found on late bill text and summaries. However, the model continues to struggle with overall generalization on unseen legislative records potentially due to temporal inconsistencies in metadata, missing grouped bill families and class imbalance.

Balanced SVM + metadata					
Jurisdiction	**Bill_Number**	**Bill_Title**	**Recorded**	**Predicted**	**Original SVM**
Alabama	H 208	State Government	Failed	Passed	Failed
Arizona	H 2175	Claims and Prior Authorization	Passed	Failed	Failed
Alaska	HCR 3	Artificial Intelligence Legislative Task Force	Pending	Failed	Failed

Table 5. Examples of 2025 validation errors for the selected and original SVM models.

Three cases show different classification challenges. Bill H208 recorded as failed was incorrectly classified as Passed by the selected model, although the original SVM correctly classified it as Failed. This demonstrates that changes in feature selection and model configuration can improve performance but still introduce errors to individual bills.

Bill H2175-Passed was incorrectly classified as Failed by both models just like HCR 3 although this second one was actually in Pending status. This highlights that with the available features, it is still very difficult for the model to distinguish among classes, regardless of the model configuration.

Information introduced via later versions of the bill in metadata, along with the possible missed interactions between related bills and potentially leaving out contextual text data could be contributing to the overall classification errors. In addition, class imbalance and differences in legislative age across years may be affecting the model ability to distinguish between pending bills and fully resolved bills.

Although these factors provide explanations for the observed errors, the individual misclassifications alone do not establish their causes. Additional investigation of bill histories, metadata availability, and related legislation would be necessary to determine their specific contributions

6. Unsupervised Learning

Nonnegative matrix factorization (NMF) and latent Dirichlet allocation (LDA) analyzed 2,567 NCSL summaries from 51 jurisdictions. Summaries reduce full-text boilerplate. NMF extracts additive TF-IDF patterns; LDA estimates probabilistic topics from word counts. Comparing them checks whether themes depend on method; full texts supported case review.

Features were lowercase unigrams/bigrams after English-stopword removal, with min_df=3 and max_df=0.85 (6,444 terms). Removing relates, provides, act and section left 6,168 TF-IDF features for refined NMF. Seed-42 fits used NNDSVDa and up to 500 iterations for NMF; batch learning and 30 iterations for LDA. Metadata were excluded.

For each topic's top 12 terms, we measured NPMI coherence (co-occurrence), diversity (unique-term share) and mean pairwise Jaccard overlap. Higher coherence/diversity and lower overlap favor coherent, distinct themes; representative-summary review checks their meaning.

Table 6. Topic-model diagnostics: 12 leading terms per topic.

Model and preprocessing	Topics	Coherence ↑	Diversity ↑	Overlap ↓
NMF: baseline	8	0.160	0.885	0.024
NMF: legal stopwords (retained)	8	0.211	0.896	0.026
NMF: legal stopwords (rule-selected k)	10	0.208	0.892	0.021
NMF: legal + geography removal	8	0.237	0.906	0.022
LDA: baseline (retained)	8	0.090	0.594	0.143
LDA: highest tested coherence	10	0.104	0.575	0.124

Single fits (seed 42), not multi-start averages. Vocabulary changes prevent isolating model effects.

With identical baseline features and eight topics, NMF had higher coherence/diversity and less overlap than LDA. Legal-word and geographic-name removal increased NMF coherence, but scores alone do not establish meaningful themes.

NMF tested 5–12 topics with five starts each (one NNDSVDa, four random). Eight was the smallest count within 0.01 of the best mean coherence (0.225 versus 0.232 at nine). Ten was more stable (0.921 versus 0.868) and won the quantitative rule; we retained eight for broader themes. LDA tested 5/8/10/12 topics: ten maximized coherence, while eight improved diversity and matched NMF's topic count. LDA seed sensitivity remains untested.

NMF identified themes including political synthetic media, automated decisions, intimate imagery, health services and algorithmic pricing (Figure 3). Topics 0 and 1 mainly contained generic AI wording.

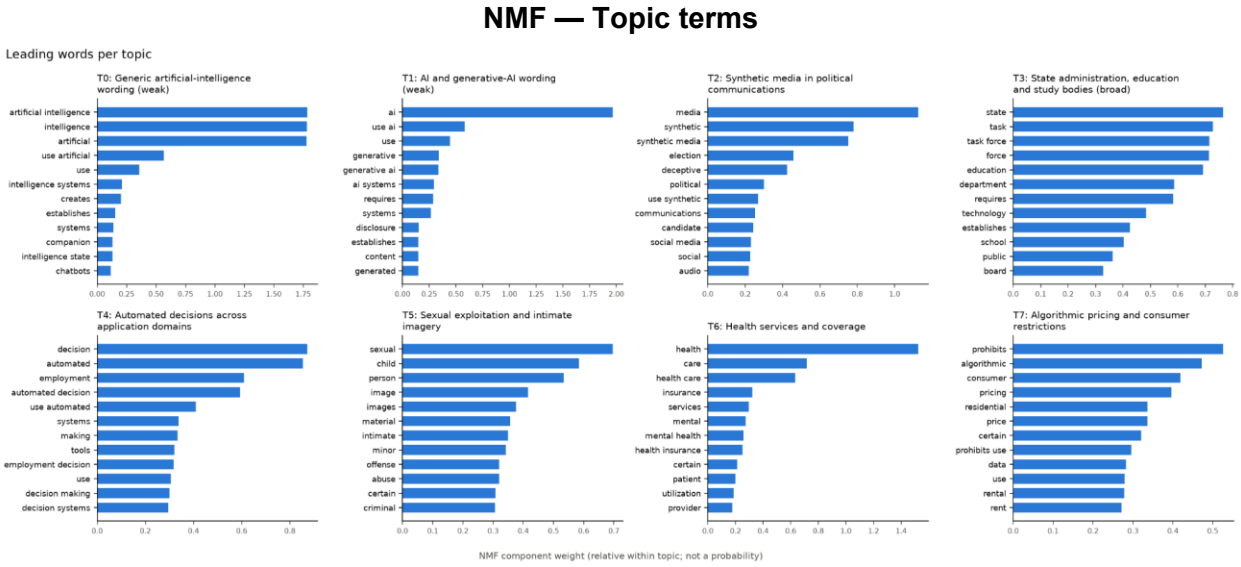


Figure 3. NMF: 12 leading terms per topic. Weights are relative within topics, not probabilities.

LDA — Topic terms

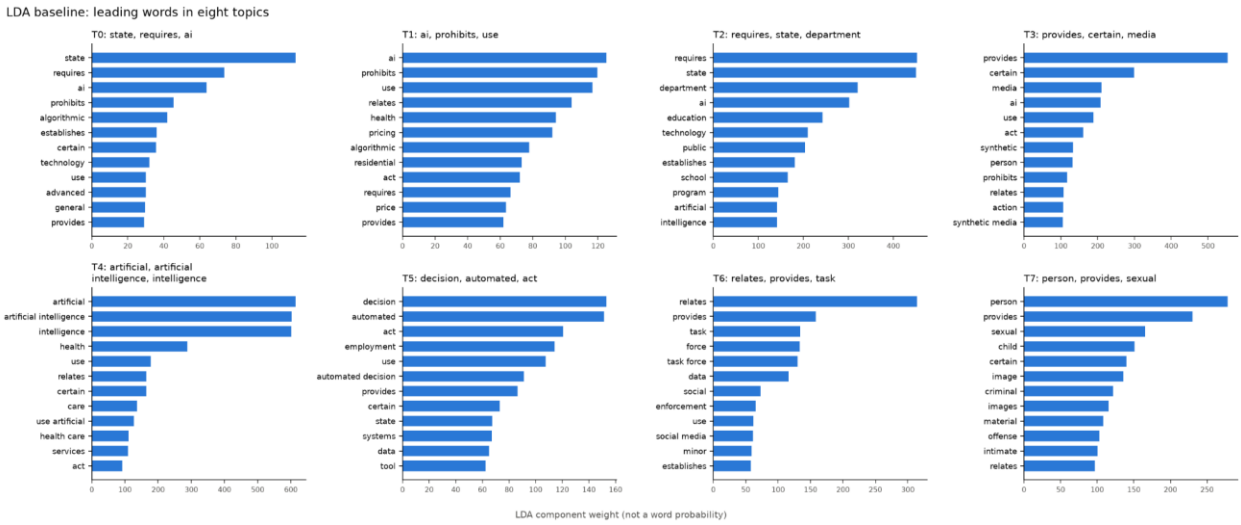


Figure 4. LDA: 12 leading terms per topic. Component weights are not word probabilities.

Figure 5 describes language patterns across states, not bill counts or policy priorities.

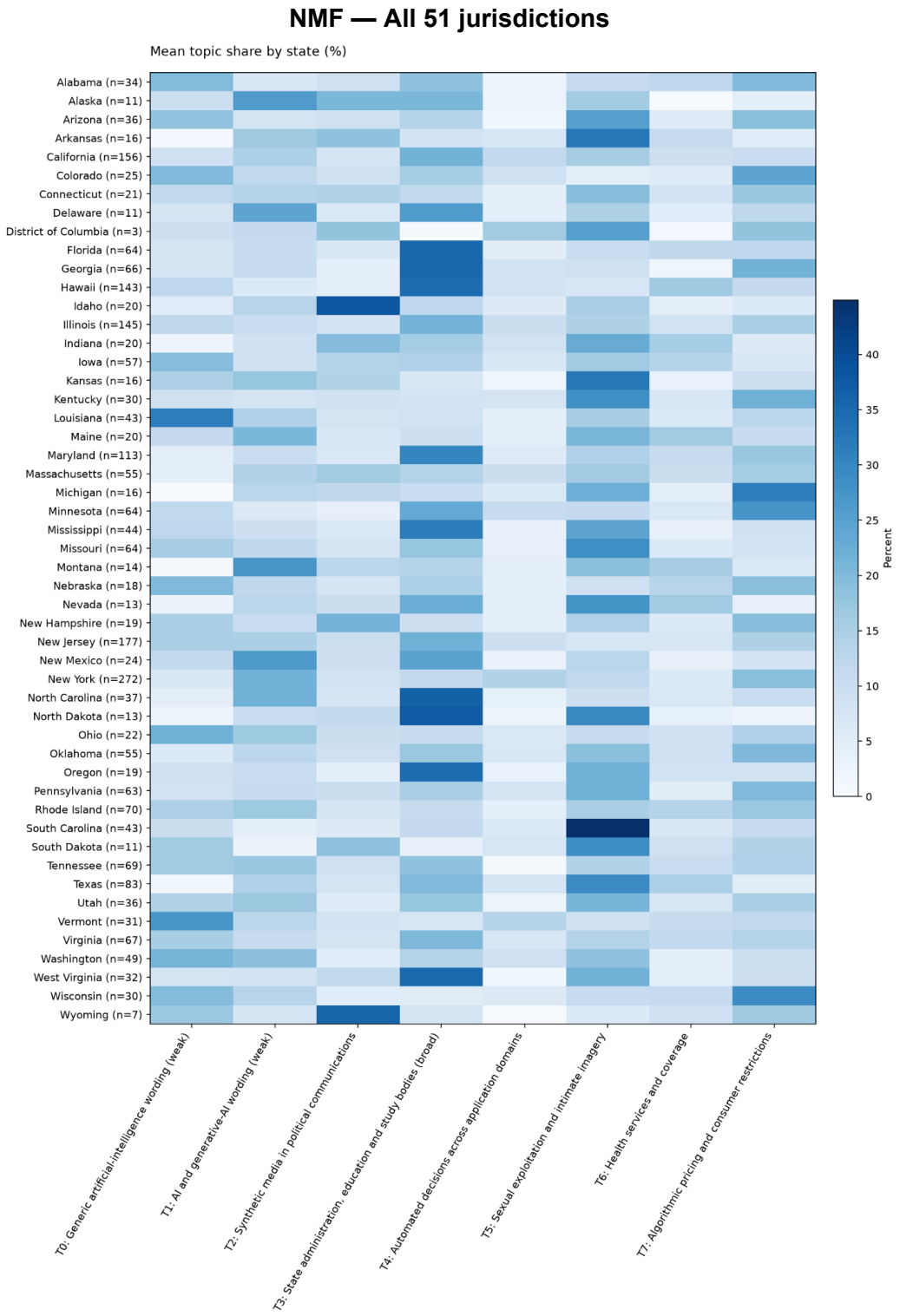


Figure 5. NMF: mean normalized topic weights across 51 jurisdictions; n is the record count.

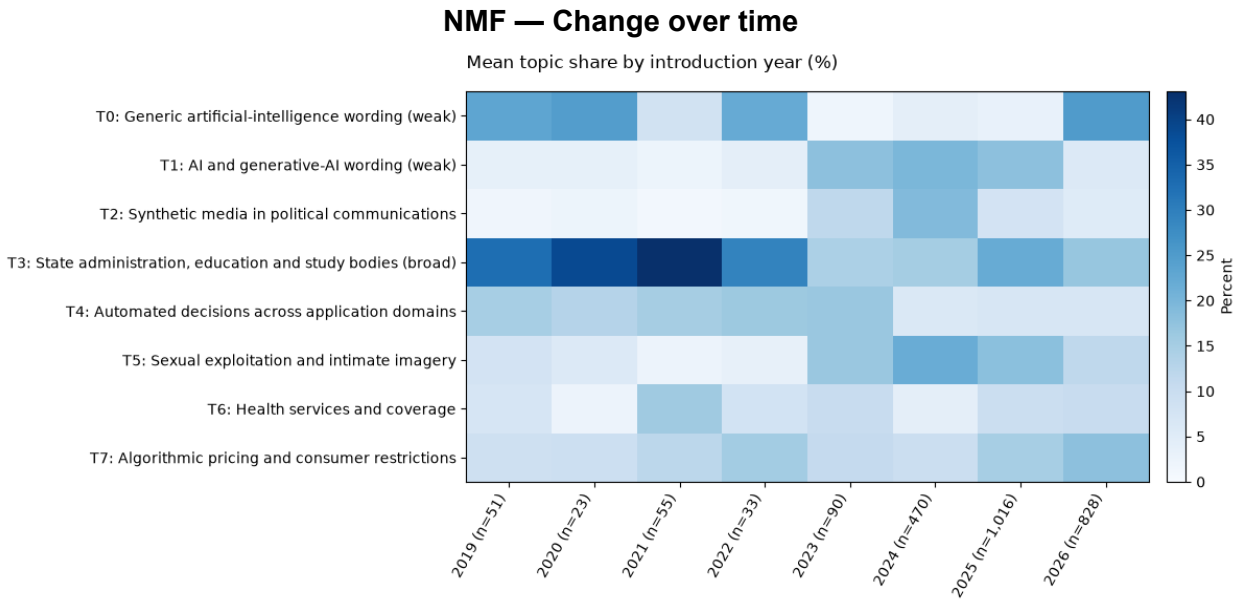


Figure 6. NMF: mean normalized topic weights by introduction year; n is the record count. The single 2018 record is omitted; 2026 is incomplete. Values are weights, not bill percentages.

Synthetic-media and intimate-imagery weights peaked in 2024; algorithmic pricing peaked in 2026 (18%). Health-services weight rose from 4% in 2024 to 10% in 2026, below 2021's 16%. The 2026 cohort ends April 2 and is incomplete.

Full-text review found human-involvement requirements in IN H 1421, NY A 9601 and SC S 443. NY S 6471-A (housing) and S 8115-B (lending) share notice/assessment duties but differ in remedies: housing permits alternative-process or accommodation requests; lending permits correction and appeal for denials based on incorrect personal information.

IL S 2481 (workforce), UT H 168 (education) and CO S 240 (criminal-justice discovery) instead create study/reporting bodies. IL S 2481 and IN H 1421 thus show different approaches within employment: institutional study versus direct duties.

The eight selected bill-pair relationships held in 96/96 NMF checks (12 runs) and 28/32 LDA checks (four runs). This measures stability, not accuracy. Notebook 03 and topic_case_review.json record selections and passages.

Figures 4 and 7 show LDA terms and state weights. Reviewing five high-weight summaries per topic found education and automated-decision themes, alongside mixed domains, boilerplate and repeated summaries. Topic numbers do not correspond to NMF's.

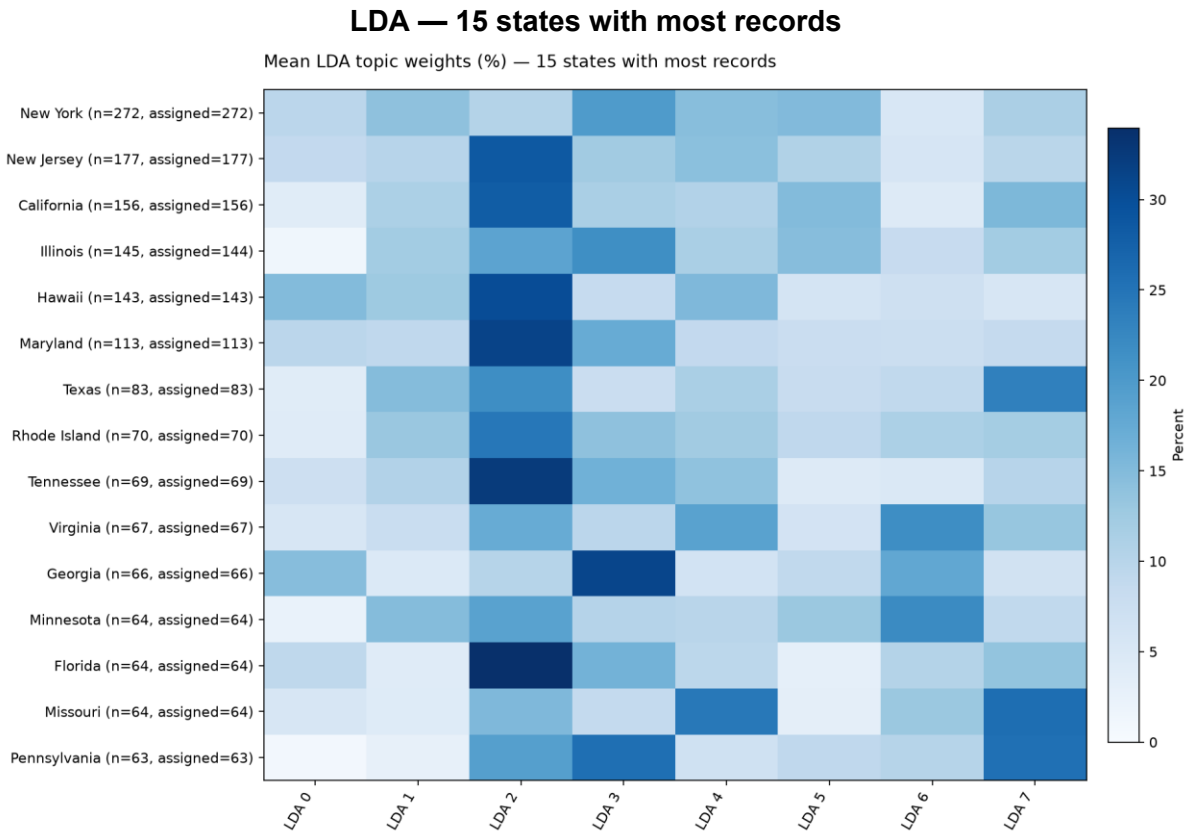


Figure 7. LDA: mean topic weights for the 15 states with most records; n is total records and assigned counts usable summaries.

7. Discussion

Part A: Recorded outcomes and source validation

The original 2026 test shows why accuracy alone can be misleading. Of 828 bills, 631 were Pending, so a model that always predicted Pending missed every Passed and Failed bill. Our SVM identified some bills in those two groups, but it also made many errors. Even in this retrospective test, those errors were too frequent for reliable use.

Figure 8 shows the selected model's errors across the three recorded outcomes. In particular, 283 Pending bills were classified as Failed.

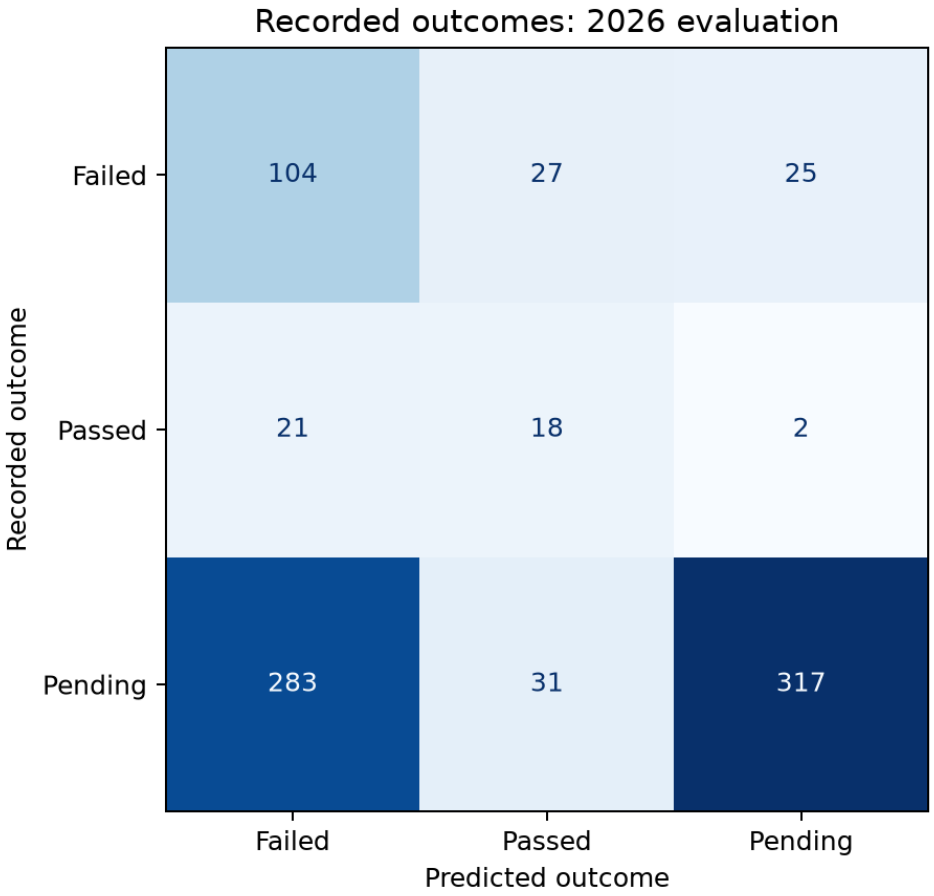


Figure 8. Recorded and predicted outcomes for the selected metadata SVM on 828 bills from 2026. Rows are recorded outcomes; columns are predictions.

We also checked whether errors in the source labels contributed to the weak results. We trained two models on the same 629 bills, one using the original labels and one using labels corrected through our source review. Both used the same settings and were scored against the same 894 bills from 2025, with the same selectively corrected evaluation labels. The corrected-label model identified 32 of 146 Passed bills, compared with 25 for the original-label model, while precision stayed around 30%. Macro F1 rose from 0.518 to 0.531 and balanced accuracy from 54.0% to 55.4%; overall accuracy fell slightly, from 62.3% to 62.0%. Balanced accuracy gives each outcome group equal weight when averaging recall. A 95% paired-bootstrap interval over evaluation states for the macro F1 difference ranged from approximately −0.003 to 0.030. Because that range includes zero, it does not establish a clear gain. It also conditions on the fitted models and does not account for uncertainty from training, source review or model selection. The corrections improve the evidence behind the labels, but their effect on performance remains uncertain.

We also asked whether adding the full text of a bill helped distinguish its recorded outcome. For this development comparison, we used the existing TF-IDF pipeline and excluded eight training texts dated after 2024 and 20 evaluation texts dated in 2026 from both models. That left 621 training bills and 874 bills in the 2025 evaluation cohort. With metadata and text, the balanced SVM achieved 64.8% accuracy and 0.588 macro F1, compared with 62.5% and 0.538 using metadata alone. Balanced accuracy increased from 56.0% to 60.7%. The text model identified 51 of 145 Passed bills, compared with 33 for metadata alone. When it predicted Passed, it was correct 47.7% of the time, compared with 32.7% for metadata alone. Both models used C=10, the same corrected labels and preprocessing fitted only on training data. Text also improved

macro F1 in the corresponding 2023 and 2024 development comparisons, but performed worse on the small 2022 cohort of 23 bills.

Removing sponsor features gave slightly lower accuracy (64.5%) and slightly higher macro F1 (0.591) on the same 2025 bills. For the reduced feature set, stronger regularization (C=0.1) lowered accuracy to 62.4%; replacing full bill text with NCSL descriptions gave 62.2%.

We conducted these exploratory comparisons after examining the original 2026 test. They do not replace that test or establish a new final model. Different row sets also prevent direct score comparisons with the original test. Earlier-year checks use retrospectively corrected snapshot labels, and text-version screening does not reconstruct what was known at introduction. Sponsor data and NCSL descriptions also lack independently reconstructed historical versions. Notebook 08 reproduces the supplementary comparisons from saved source observations.

Several limits remain, including the use of retrospective inputs and outcomes we could not verify. Our larger review independently resolved outcomes for 311 of the 1,523 matched bills; the other 1,212 kept source labels that we could not independently verify. This selective review cannot establish label accuracy across the full corpus. A Pending label means the outcome was unresolved in the source snapshot, not that the bill eventually failed. Related bills may also share wording across years even after exact duplicates are removed. A sensitivity check removed training texts highly similar to evaluation texts, but the main chronological split did not separate bill families. A stronger follow-up study could use a common outcome horizon, reconstruct what was known at that point and keep related bill families together during evaluation.

Part B: Topic analysis

Unexpectedly, clearer topics did not consistently improve classification. In the matched metadata-SVM comparison (Notebook 06), NMF features raised 2025 macro F1 from 0.570 to 0.574 but reduced the historical mean from 0.486 to 0.405. Vocabulary and topics were fitted within each training split. This suggests that interpretable themes alone do not guarantee useful predictive information across years.

Boilerplate and short summaries made topics difficult to distinguish. Removing four common legal terms improved NMF coherence, while selected full-text reviews clarified differences in legal duties that similar summaries could obscure. Two topics remained generic, so quantitative improvements did not fully resolve interpretation problems.

With more time, we would sample additional bills independently of topic weights, use a second reviewer to assess interpretations, and test LDA with NMF's stopwords across multiple seeds. This would help separate preprocessing effects from model differences. The eight selected cases remain illustrative rather than representative.

8. Ethical Considerations

Part A: Data and outcome classification

Legislative text and sponsor affiliations are public, but a model can still give a misleading picture of a bill. Passed, Failed and Pending describe recorded legislative states, not a proposal's quality or social value. Party affiliation and political control may reflect broader institutional patterns, but our results do not show that they cause a bill to succeed or fail. Given the model's errors, its classifications should not replace official bill-status checks or guide legal compliance decisions.

The corpus reflects the measures collected and the text available in its published snapshot. Missing summaries, changing sponsor rosters and uneven legislative follow-up can affect which records are usable and how they are interpreted. Publisher-generated policy categories are distinct from verified human judgments. We preserve the original source, link prepared rows to

it, and keep dated source observations separate from baseline labels. We leave ambiguous companion-bill histories unresolved rather than assigning unsupported outcomes. Public access to a source does not imply unrestricted reuse of every linked document or API service.

Part B: Topic analysis

Topic labels can oversimplify policy debates. Short summaries and repeated wording may obscure differences in legal duties, affected groups and remedies. We use labels as interpretive aids and check selected cases against the relevant full-text version. A low topic weight does not imply that a policy concern or affected group is unimportant.

State and year averages describe language in the collected summaries, not policy commitment or implementation. Uneven record counts and incomplete 2026 coverage can distort comparisons. We report record counts and coverage limits rather than rank jurisdictions or infer an absence of regulation.

Selecting eight cases after viewing results risks confirmation bias. We document selection reasons, supporting passages and differences between bills, and restrict conclusions to the reviewed evidence. Sensitivity checks measure consistency, not correctness; these cases are not independent validation. Broader claims would require a separately sampled set and independent review.

9. Statement of Work

Josue Nunez: Developed the shared preparation workflow and data-quality checks. Reviewed source dates, sponsor differences and recorded outcomes, integrated repository changes, and checked reproducibility. Extended the common classifier comparisons, error analysis and final evaluation, and ran supplementary pre-2026 label and text comparisons. Contributed the introduction, related work, data overview, source-validation discussion and related ethical considerations to the report.

David Hernandez Montes: Coordinate supervised model comparisons, feature engineering, evaluation, and writing of feature engineering and supervised learning sections of the report.

Meng Chieh Chiang: Developed the initial topic-analysis notebook and contributed to interpreting the themes.

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Project Links

GitHub repository: https://github.com/dhMontes/siads696-state-of-ai

Google Doc (Comments Enabled): SIADS 696: Final Report - State AI Policy